

When Financial Agents Code: Auditing Spurious Alpha, Data Leakage, and Execution Errors with Guarded Domain Libraries

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Abstract

Financial agents increasingly write the quantitative code behind their conclusions, yet a plausible backtest can conceal future information, a misapplied corporate action, or a validation claim that never executed. We present FIN-SKILLS, a financial library of 129 skills, 48 algorithm-backend catalog entries, and 36 executable research guards that turn dated domain conventions into structured computation and artifact-bound audit records. Across three seeded worlds, the guard suite detects all 36 of 36 planted defects with no false alarms on the clean references, and 13 of 13 numerical checks meet their stated tolerances. In a frozen paired study of first-submission code generation with two open code models, none of 48 programs meets the pre-registered validity criteria: 37 responses break the code-only format and every executed program fails numerical and split checks. With the first code fence extracted post hoc, 37 programs execute but only 1 is valid; 24 programs mishandle cumulative split factors and 15 use same-session or future prices; documentation alone produced no library calls. In a frozen follow-up with three attempts, enforcing lag and split checks cut accepted-but-wrong programs from 22 of 24 to 2, and to 0 with an enforced library call, but did not raise valid programs (1, 1, and 0 of 24). The checks prevent silent errors from being delivered; they do not make small models write correct code. HiSTrim and persistent memory are optional integrations whose results are reported separately.

1 Introduction

A financial agent can produce a convincing research report while joining a disclosure to dates before publication, treating a stock split as a return, or reporting a check that never executed. These failures can create apparent alpha without a usable investment signal, and they increasingly arise in code the agent writes itself. Prevent-

ing them requires financial conventions to be expressed as executable input requirements and tests, alongside the algorithms that consume the data.

FIN-SKILLS provides this infrastructure. Its domain skills identify appropriate libraries and dated failure modes; its numerical interfaces expose parameter and data contracts; and its research guards check availability times, corporate actions, validation splits, costs, and other declared assumptions. Structured findings distinguish a detected defect from a missing input or a successful check. Artifact-bound execution records allow a controller to enforce those checks rather than trust an agent’s prose.

We ask whether language models can reliably build quantitative computations without such a library, and what the library’s checks and primitives establish. E1 measures whether the guard suite detects known defects without flagging clean references. E2 measures whether agents discover and execute the interfaces and whether submitted artifacts meet the audit contract. E3 compares numerical primitives with independent references. E4 freezes a paired code-generation study in which two open code models implement four allocation and signal rules with and without the library’s documentation, graded by an independent executable oracle, then repeats the tasks with three attempts while a controller enforces the library call and lag and split checks. First submissions are almost never valid, and the dominant errors (re-cumulated split factors and same-session prices) execute without raising an error. Enforcement stops most wrong programs from being accepted but does not produce more correct ones, which separates the library’s auditing value from any claim of better generation.

The distinction matters for interpreting an earlier portfolio experiment. That experiment exposed a price-compatible subset of algorithms on one short, anonymized currency path, with-

out mandatory guard enforcement. It is an exploratory component ablation of algorithm access, not a comparison of the complete guarded library against an unguarded system. We retain its returns and execution traces, but do not use its losses to infer either that financial tools are ineffective or that memory would repair those losses.

Our contributions are (1) a library of 129 financial skills, 48 algorithm-backend catalog entries, and 36 executable research guards; (2) structured computation and artifact-bound verification interfaces connecting financial assumptions to actual execution; (3) evidence on planted-defect detection, execution behavior, and numerical parity; and (4) frozen, hash-bound audits of financial code generation that separate format failure, silent financial error, and library uptake, and measure what enforced checks change: fewer wrong programs accepted, not more correct programs written. HiSTrim and persistent memory provide optional context and experience services; their auxiliary studies do not determine the core contribution.

2 Related work

FinAgent combines multimodal financial information with trading tools, reflection, and memory (Zhang et al., 2024); StockBench evaluates sequential portfolio decisions using prices, fundamentals, and news (Chen et al., 2025). The Finance Agent Benchmark evaluates financial research tasks (Bigeard et al., 2025). These works study tool use and memory within language-model agents. Our contribution supplies the financial knowledge, numerical contracts, and executable research checks that such agents can reuse. A separate exploratory comparison varies algorithm access under identical data.

Behavioral testing offers a structured view of model capabilities beyond aggregate scores (Ribeiro et al., 2020). Financial backtest overfitting and selection bias (Bailey and López de Prado, 2014; Bailey et al., 2017), alongside look-ahead leakage in financial agents (Li et al., 2025), motivate explicit checks on decision evidence. The persistent memory component adapts the mushroom-body model of Huang et al. (2024) to update state from delayed financial returns, linking within-task attention dynamics with cross-episode retention.

3 Library architecture: knowledge, computation, and verification

FIN-SKILLS connects three reusable layers. Skills describe financial assumptions and library conventions. Numerical interfaces execute methods against declared inputs. Research guards return structured findings that a controller can require before accepting an artifact. A successful algorithm call alone does not imply that the required audit ran.

Guarded research workflow. For a task and its available data, an agent selects a skill, inspects a numerical interface, and proposes an artifact. A controller executes the relevant checks on that artifact. Failures return actionable findings; missing required evidence remains incomplete. The controller accepts only artifacts meeting its declared policy and saves the associated execution records. This workflow is distinct from offering an agent optional tool access and trusting its claim to have validated the output.

Optional experience services. HiSTrim operates within a single session by evicting tokens from the KV cache during generation. Persistent memory operates across sessions by storing parameter updates in an external module. The two mechanisms operate on different timescales: context pruning controls what the model attends to right now, whereas persistent memory retains summary statistics of what happened in previous episodes. Neither service is required to run the library’s guards or reproduce its numerical primitives.

Grounded feedback updates. When an episode concludes, the agent records the market context, the chosen tool, and the realized return. Only verified, settlement-cleared outcomes update memory. Negative returns are retained as learning signals, while trades relying on look-ahead data or unadjusted prices are rejected by guards to prevent memory corruption.

4 Professional tools and domain knowledge

FIN-SKILLS exposes financial methods alongside the information an agent needs to run them: task scopes, input requirements, dependencies, parameter conventions, and dated source verifications. Domain skills route requests to relevant libraries;

Component	Role in research workflow	Target question	Implementation and validation
Financial skills	Route tasks and document dated conventions	What assumptions and failure modes apply?	129 skill documents
Numerical interfaces	Discover and execute quantitative methods	Do selected computations agree with references?	48 catalog entries; E3 tests eight primitives
Executable guards	Diagnose data, accounting, and validation defects	Which required checks passed, failed, or lack inputs?	36 registered guards; E1 exercises 13
Execution records	Bind verification to the submitted artifacts	Did the required checks actually run on this code and data?	Explicit pass, fail, and incomplete states
Optional extensions	Select context and retain verified experience	How do auxiliary context and memory services affect tasks?	HiSTrim and persistent memory; separate studies

Table 1: Core financial library components and optional context and memory integrations.

per-library skills provide implementation guidance. The algorithm catalog covers portfolio construction, forecasting, volatility estimation, technical signals, and execution schedules, providing uniform programmatic contracts over quantitative methods.

Agent-selected execution. An agent can discover candidate algorithms, inspect their requirements, and execute them with custom parameters. Structured outputs separate numerical results from missing dependencies or runtime errors. An optional strategy adviser provides rule-based suggestions without overriding model choices. In our evaluation, agents access 18 price-compatible algorithms, with requests and executions logged independently from free-form text.

Information and model interfaces. Retrieval interfaces chunk packaged skills and documents, returning relevant passages under strict as-of cut-offs to prevent look-ahead bias. Forecasting and prediction adapters provide a unified fit–predict lifecycle, while portfolio optimizers expose specialized execution interfaces. Appendix B details the complete interface inventory, model lifecycle, and retrieval pipeline.

5 Execution and feedback verification

Tool execution and memory updates require verifiable records of what actually ran. The execution controller computes cryptographic hashes over input data, executed code, and returned allocations. If an agent modifies an artifact after execution, previous validation receipts become invalid. The controller returns `PASS` if all required guards pass, `FAIL` if a guard detects a defect, and `INCOMPLETE` if evidence is missing. This mechanism enforces execution provenance without modifying the underlying algorithm.

For cross-episode memory, timing and accounting checks verify that delayed outcomes cleared settlement before they update persistent state. Trades that rely on look-ahead data or unadjusted corporate splits are blocked, preventing corrupted feedback from distorting the agent’s memory. Unsettled trades wait for settlement, while valid unprofitable trades remain eligible learning signals.

The audit study evaluates models under unconstrained generation, optional self-checking, and mandatory verification. Appendix A describes the evaluation setup, defect detection benchmarks, and empirical audit results across model tiers.

6 Library evaluation: defects, execution, and numerical parity

We evaluate the financial library at three boundaries: detecting invalid research inputs, executing declared interfaces, and reproducing numerical references; Section 7 adds the code-generation audit (E4). Table 2 summarizes the observed evidence. Each row has its own unit and denominator.

E1: detecting research defects. The benchmark plants 12 defect families, including temporal leakage, unadjusted corporate actions, and universe-selection errors, in three synthetic worlds. The suite records 36/36 detections and no false alarms on its clean references. This is suite-level recall: a defect may be caught by another applicable guard even when one intended check misses it. Repeating a family across seeds does not create a new defect type. Appendix A describes the independent oracle and the distinction between diagnosis and agent repair.

E2: execution and audit compliance. We count actual dispatches and final artifacts rather than statements that a check ran. The exploratory

Study	Endpoint	Observed result	Scope
E1: guards	Planted-defect detection	36/36 detected (100%); no clean-reference false alarms	12 defect families in three seeded worlds; 13 core guards exercised
E2: execution	Valid decisions and successful tool requests	64/64 valid decisions across four conditions; 43 successful autonomous tool requests	16 autonomous decisions; requests include discovery, reading, advice, and computation
E2: audit	Artifact validation and reporting fidelity	Reference submission accepted; pilot Opus Sharpe gap 0.188 to 0.034	Historical pilot; self-reported guard use is not an execution receipt
E3: parity	Agreement with numerical references	13/13 checks within case-specific tolerances	Selected primitives; maximum absolute error 1.18×10^{-8}
E4: code	Valid programs; accepted but wrong	First submission 0/48 (1/48 post hoc). Enforced, three tries: valid 1, 1, 0 of 24; accepted but wrong 22, 2, 0	Two model sizes, four rules, three seeds; arms: raw, raw with checks, library with checks

Table 2: Core library evidence. Detection, executable completion, audit acceptance, and numerical agreement are different endpoints. No row establishes universal correctness or profitable trading.

portfolio study completed all 64 decisions, including 16 with autonomous tool access, and logged 43 successful requests in that arm. These requests include catalog lookup and guidance retrieval as well as numerical algorithms. In the separate audit feasibility study, optional guards and mandatory enforcement have different acceptance outcomes. Mandatory rejection protects the submission boundary but does not constitute successful repair: the 14B model reached its turn limit in all three mandatory cells. The reference submission passed the public checks and independent grading.

E3: numerical parity. Eight checks compare equal weighting, inverse-volatility weighting, minimum-variance optimization, normal and historical VaR/ES, historical volatility, fractional-difference weights, and an augmented Dickey-Fuller statistic with closed forms or independent NumPy, SciPy, pandas, and statsmodels references. All meet their recorded tolerances; the largest absolute difference is 1.18×10^{-8} for the optimization weights. The optimization case allows 10^{-3} while other cases have their own tighter tolerances. This supports agreement for the tested calculations, not exact equality or validation of all catalog entries.

Auxiliary integrations. The context and memory studies test optional consumers of the library’s results. Their methods and outcomes appear in the appendices. The public FinQA RAG and reranking runs retain their original official scores (4/192 correct programs across six model and retrieval cells). A post-hoc reanalysis bound its target-blind extractions by hash before grading: accepting a single fenced answer adds no correct pro-

gram, and counting explicit correct final numbers raises the total to 7/192. That second count is not program accuracy. Of the 96 Mistral answers, 75 have no final answer, so format repair alone cannot isolate retrieval quality (Appendix B). Interface corrections require a new versioned comparison rather than retrospective replacement of the recorded grades.

7 E4: Financial code generation with and without enforcement

Can an open code model write a correct allocation or signal rule from a specification alone, and does the library’s documentation, or enforcing its use and checks, change that? We froze a paired first-submission study before model inference. Qwen2.5-Coder 7B and 14B each wrote one program for four rules (diagonal risk parity by inverse volatility, positive cross-sectional momentum, a 5/20 moving-average crossover, and volatility-targeted momentum), three data seeds, and two arms: 48 programs. Both arms received identical raw quotes with split jumps, cumulative split factors known on their event dates, the mathematical rule, the one-bar timing rule, one response of at most 2,048 tokens, and no execution feedback. The library arm additionally received the actual catalog output and interface and guard documentation (mean prompt 11,667 versus 10,756 tokens). Programs ran in the Landlock/seccomp worker. An independent NumPy grader, shared by both arms, requires execution, agreement with the reference within 10^{-8} , unchanged positions through session k when prices at and after k are perturbed ($k = 32, 64$), and identical output for

the economically equivalent split-adjusted input. A program is valid only if it passes all.

Pre-registered outcome. No program was valid in either arm (0/48). The protocol asked for Python only, optionally inside one fence. In all, 37 responses appended an explanation after the fence (24 from 7B and 13 from 14B), so the frozen parser compiled prose and recorded a `SyntaxError`. The 11 executed programs all failed numerical agreement and split invariance. No response was truncated. None of the 24 library-arm responses imported or mentioned the library, and no algorithm or guard call was traced. The comparison therefore measures documentation exposure without uptake; it does not estimate the effect of using FIN-SKILLS.

Post-hoc extraction and failure modes. To separate formatting from computation, we froze an arm-blind extraction of the first complete fence by hash before execution and reran the same worker, sandbox, and grader on another Linux host; it reproduced all 11 official executions exactly. Then 37 programs execute, but only 1 of 48 is valid. The failures are silent in the sense that matters for research: 24 of 24 14B programs, in both arms, applied a cumulative product to split factors that the prompt defines as already cumulative, and 26 executed programs change their positions when the same prices are presented split-adjusted. In 15 executed programs a position moves when prices at or after its own session change; 12 use the same-session close and 3 use later data. The 7B programs that still fail raise shape, index, or pandas-API errors.

What the library guard observed. Run post hoc on each executed program, the library’s causality guard flagged all 3 programs with strictly-future dependence and none of the 22 programs that passed the grader’s timing probes. It passed the 12 same-session violations: the guard asks whether values before k depend on data at or after k , which is feature causality, whereas a position must also exclude session k itself. The execution-lag contract therefore needs its own check rather than an inference from a passed causality test. Together with E2, these results show that raw first submissions are unreliable in ways that still execute, and that optional documentation is not used.

Enforced library calls and checks. We therefore froze a follow-up on the same rules, seeds, models, prompt packet, and hidden grader, with up to three attempts and controller feedback in three arms of 24 episodes: coding from scratch with execution feedback only; coding from scratch with enforced lag and split checks; and the library documentation with an enforced `fin_skills.algorithms.run` call plus the same checks. The lag check applies the library’s causality guard to positions shifted up one row ($k = 40, 72$), closing the same-session gap above; the split check requires identical output for the split-adjusted representation. Feedback never reveals reference weights, and every arm parses the first complete fence. A deliverable counts only if the controller accepts it and the unchanged grader finds it valid.

Enforcement did not produce more valid programs (Table 4): 1, 1, and 0 of 24. It changed what is delivered. With execution feedback alone, 23 programs were accepted and 22 of them were wrong (91.7%), including 12 split and 9 timing failures. With enforced checks, accepted-but-wrong programs fell to 2 (8.3%), both of which pass the grader’s timing and split probes and fail only numerical agreement; with the library requirement, none was accepted. The library was hard to use from its interface text: 42 of 72 library-arm attempts reference FIN-SKILLS, 8 trace a call, and 51 fail to run, mostly on output shape, nonfinite values, and argument errors. In 7 attempts the model followed a parameter name in the supplied interface text that the function does not accept; all 7 also re-cumulated split factors and would have failed the split check. For these models and this budget, the checks turn silent errors into explicit rejections; correct construction still needs a stronger model, more repair budget, or a simpler interface.

8 Component ablation: algorithm access without mandatory guards

This exploratory study tests a restricted algorithm-access component of the library. The dispatcher offers discovery, skill reading, computation, and optional advice, but does not enforce the library’s research guards on final allocations. We treat it as an unguarded component ablation, not as a test of the complete guarded workflow. No matched guarded arm was run in this experiment.

Model	Arm	Executed		Post-hoc extraction				Mean per response	
		Official	Post-hoc	Correct	Split fail	Timing fail	Valid	Output tok.	Gen. s
7B	Raw	0/12	7/12	1	2	2	1/12	398	8.4
7B	Library	0/12	6/12	0	0	4	0/12	337	7.8
14B	Raw	5/12	12/12	0	12	6	0/12	312	9.4
14B	Library	6/12	12/12	0	12	3	0/12	286	9.0

Table 3: Paired first-submission code generation. Under the pre-registered parser no program is valid in any cell (0/48). Post-hoc columns take the first complete code fence; failure counts are among executed programs. No library-arm program called a library algorithm or guard.

Arm	Valid	Acc.	Wrong	Lib.	Tries
Raw	1/24	23	22	—	1.25
Raw + checks	1/24	3	2	—	2.79
Lib. + checks	0/24	0	0	5/24	3.00

Table 4: Enforced follow-up (job 1765432; 7B and 14B pooled, up to three attempts). Raw receives execution feedback only. Valid: accepted and valid under the hidden grader. Acc.: accepted. Wrong: accepted but invalid. Lib.: episodes with a traced library call in any attempt. Tries: mean attempts.

8.1 Experimental setup

We test Qwen2.5-Coder 7B and 14B under four conditions: historical prices alone; prices with static skill excerpts; prices with precomputed tool outputs; and autonomous access to the library’s algorithm catalog. Each condition runs across inference seeds 11 and 23, with a budget of five interaction turns per decision (capped at 256 generated tokens per response). Conditions run in randomized order.

In the autonomous condition, the agent receives descriptions of available tools. It can inspect algorithm specifications, run an algorithm with custom arguments, or query a rule-based strategy adviser. Tool calls are dispatched against trailing market snapshots and return structured numerical results. In our evaluation, agents access 18 price-compatible quantitative routines.

A decision without valid portfolio weights leaves existing allocations unchanged. Requests, outputs, and intermediate reasoning traces are logged separately, and final weights are frozen before scoring.

8.2 Data and evaluation protocol

We evaluate on daily exchange rates for EUR against USD, JPY, GBP, and CHF from ECB reference data. Messages use anonymized asset identifiers and relative indices rather than calendar

Condition	7B	14B
No library	+0.100	−3.324
Skill text	−2.930	−4.567
Fixed tool outputs	−3.075	−3.030
Autonomous library	−3.984	−3.783

Table 5: Mean cumulative proxy return (%) across inference seeds with 5 bps transaction costs per side.

dates.

Each decision observes 126 trailing days. The evaluation spans the first 84 trading days of 2025 (ending May 2), with four rebalancing points spaced 21 days apart. Decisions at step t execute at day $t + 1$ prices to prevent look-ahead bias. Portfolios are long-only without leverage, with unallocated capital held in cash. We charge 5 basis points per side for rebalancing and final liquidation. Cash, equal-weight buy-and-hold, and deterministic rules serve as reference baselines.

8.3 Observed execution and returns

Across 16 trials, models generated 64 valid final portfolio allocations. The autonomous conditions account for 16 decisions and 43 successful requests: 13 from 7B and 30 from 14B. Requests include catalog discovery, skill reading, strategy advice, and algorithm execution; they are not 43 numerical algorithm executions. The completed autonomous conditions recorded no tool or format errors. An earlier parser-failure attempt is retained separately and is not included in these completed trials.

Tool choices diverged across model scales: 7B selected cross-sectional momentum, Bollinger reversion, and volatility-targeted momentum, whereas 14B predominantly used Bollinger reversion alongside trend-following indicators.

Table 5 shows returns across all arms. While autonomous 14B improved over its text-only condition by 0.783 percentage points, neither autonomous model beat the unassisted baseline

(+0.100% for 7B, −3.324% for 14B).

Recorded portfolio holdings show different exposure choices across conditions: mean risky asset exposure for 7B rose from 0.500 without the library to 0.969 with autonomous tools; for 14B, exposure shifted from 1.000 (all-in) to 0.500. Autonomous runs used 38 model calls per model across eight decisions, totaling 1,155,782 prompt tokens and 10,165 completion tokens.

The valid decisions and successful requests provide interface evidence for E2. The returns describe one short, anonymized currency path with transaction costs. They do not establish that the financial library is ineffective, nor that missing guard enforcement caused the losses: the experiment lacks the paired intervention required for either conclusion. A full-workflow comparison must hold data, models, budgets, execution rules, and costs fixed while varying mandatory guards. Memory and context pruning are separate interventions and are not inferred to remedy these returns.

9 Discussion: financial reliability as an executable contract

The strongest evidence for FIN-SKILLS concerns defects that its checks can observe and computations that can be independently reproduced. E1 establishes detection on the fixed planted families; E3 establishes numerical agreement for the tested primitives. E2 shows why optional access and enforced verification must be distinguished: an agent may describe a check without executing it, while mandatory verification may reject a submission that the agent cannot repair within its budget. Detection, acceptance, and task success must therefore be reported separately.

E4 sharpens both points. Unassisted first submissions from open code models failed in the ways a backtest would not reveal: split conventions, the execution lag, and output contracts. Supplying documentation is not the same as supplying the library, because no program used it, and a guard certifies only its stated contract: the causality check caught every strictly-future dependence but not a same-session lag until it was composed with a one-row shift. Enforced, these checks reduced accepted-but-wrong programs from 91.7% to 8.3% of episodes, yet no arm produced more valid programs, and enforced library use mostly failed on interface errors. The library’s mea-

sured value here is refusing wrong results; making agents use its algorithms correctly requires a clearer calling interface and stronger models.

The portfolio component ablation leaves the causal contribution of guard enforcement unmeasured. Exposure choices, market direction, and transaction costs all affect its returns; the saved losses do not identify which factor caused the difference between conditions. A matched guarded arm is needed to measure what enforcement changes.

Context pruning and persistent memory can consume the library’s verified records, but their gains are separate empirical questions. In the auxiliary report-QA studies, a post-hoc reanalysis keeps the official program scores and shows that relaxing fence parsing adds no correct program; missing final answers, not formatting alone, limit those runs. The current context-pruning study shows quality loss. These findings motivate interface and retention checks, not a replacement of the library’s contribution with an unproven memory benefit.

10 Conclusion

FIN-SKILLS makes professional financial guidance executable through quantitative interfaces and research guards. Its fixed defect benchmark records 36/36 detections with no clean-reference false alarms, and 13/13 numerical checks pass their stated tolerances. A frozen code-generation audit finds 0/48 valid first submissions under its pre-registered parser and 1/48 after post-hoc extraction, with silent split and timing errors dominating the executable programs. Enforcing the library’s checks cut accepted-but-wrong programs from 22 of 24 to 2 without raising valid ones. These results support reusable infrastructure that refuses spurious research findings and reproduces computations; they do not show that the library makes small models write correct financial code.

Limitations

The defect benchmark repeats fixed synthetic families across seeds; its 100% recall is not a universal leakage-detection guarantee. Numerical parity covers eight primitives, not every backend. The code-generation study uses author-designed synthetic tasks, two model sizes from one family, and three seeds on four fixed rules, which are not twelve independently sampled problems; training

contamination is not ruled out. Its fence extraction is a post-hoc sensitivity analysis run on another host. The enforced follow-up reuses these tasks after seeing E4’s failures, allows only three attempts, and its interface text named one argument differently from the library function. Passing sampled perturbations does not certify universal causality. The currency component ablation uses daily reference prices, two Qwen model sizes, and one short path, without a matched mandatory-guard arm or executable quotes and carry. Auxiliary QA comparisons use public, development-exposed, overlapping tasks and retain format and runtime failures; their format reanalysis is post hoc. Research guards enforce specified checks, not regulatory certification, complete adversarial isolation, or investment profitability. Human workflow efficiency and independent prospective validation remain unmeasured.

Ethics and Reproducibility

All models, tools, and evaluation benchmarks adhere to open, reproducible protocols. Software environments, execution digests, and audit traces are publicly available to facilitate independent verification.

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A Supporting audit and validation studies

This appendix details the execution and feedback validation checks, defect recall benchmarks, and empirical audits across model tiers.

A.1 Independent evaluation environment

We evaluate agent portfolio decisions in a synthetic market environment with full ground-truth transparency, providing quoted prices, corporate split actions, delisting events, and fundamental disclosures. A submitted strategy script returns portfolio weights, alongside a written report stating its expected Sharpe ratio. Session- t portfolio weights must use only information available by the close of session $t - 1$.

An independent oracle evaluator verifies submissions by perturbing future data, checking for post-delisting exposure, and recalculating performance under fixed transaction costs. Oracle grades are generated after the submission receipt is saved and are never exposed to the agent during its decision turns.

The oracle tests whether allocations depend on unavailable information. The recorded latent-drift reference earns a sample net Sharpe of 1.63 with a one-day lag; this is a reference strategy, not a proved ceiling on attainable performance. Wrong-side joins can inflate reported performance, but a high Sharpe alone does not diagnose leakage. Dependency perturbations and availability-time checks establish the defect.

A.2 Validation benchmarks

Defect detection benchmark. We evaluate 13 core research guards against 12 planted quantitative defect families, including look-ahead feature joins, unadjusted stock split prices, future earnings disclosure leaks, and survivorship bias. Across three seeded synthetic market worlds, the guards achieve 100% recall (36/36 planted defects caught) with zero false alarms on clean reference code.

Reporting discrepancy and compliance paradox. Table 6 measures reporting discrepancy: the absolute difference between the Sharpe ratio claimed by the agent in its written report and the ground-truth Sharpe ratio computed by the oracle evaluator.

Without tools, Opus exhibited a reporting gap of 0.188, which decreased to 0.034 when given

Model and condition	Mean absolute Sharpe gap
Opus, no library	0.188
Opus, with library	0.034
Haiku, no library	0.574
Haiku, with library	11.949

Table 6: Reporting discrepancy between agent-claimed Sharpe ratios and ground-truth oracle evaluations.

execution guards (81.9% reduction). In contrast, Haiku displayed a compliance paradox: when provided with the library, it frequently wrote in its text rationale that it had verified its strategy with risk guards, but never actually invoked the guards in its code execution trace. This superficial compliance inflated its discrepancy to 11.949, demonstrating that smaller models require automated code execution verification rather than relying on self-reported compliance.

Social sentiment and multi-source corpus verification. Across our consolidated 4,233,499-record (4.23M, 1.25 GB) cross-market financial corpus (`shards_v4`: 2,272,556 EastMoney Guba posts across 464 tickers, 1,544,478 A-share corporate disclosures across 5,122 tickers, 281,981 Xueqiu longitudinal posts across 1,004 tickers, and 134,484 US StockTwits messages across 81 tickers) and 2,980 verified KOL profiles (1,445 CN, 1,535 US; 35,772 timestamped predictions over 799 trading dates), naive follower-weighted sentiment exhibited negative out-of-sample rank correlation with subsequent asset returns (Rank IC = -0.0115): blindly following raw social volume lost money. By auditing each account’s strictly lagged prediction accuracy and applying point-in-time Bayesian credibility gating (including sign-flipping persistently anti-predictive accounts), out-of-sample Daily Rank IC flipped to $+0.0273$ ($\Delta = +0.0388$, block-bootstrap 95% CI $[+0.0295, +0.0480]$) and reached $+0.0327$ under 60-day rolling adaptive recalibration. This confirms that unstructured text signals require empirical track-record verification before driving investment decisions.

A.3 Four-condition audit feasibility

We evaluated models under four audit conditions: unassisted baseline (C0), text guidance only (C1), optional executable guards (C2), and mandatory automated audit (C3).

Operational findings. Table 7 presents results for Qwen2.5-Coder 14B under automated audit constraints. The acceptance rate was 67% under C0 and C1, 100% under C2 (optional guards), and 0% under C3 (mandatory audit), where runs reached the turn limit while attempting to fix all detected code defects. Mean token consumption under C3 was 47,491 compared to 13,877 under C2. This records additional work without successful acceptance; token consumption alone does not establish better debugging.

A reference quantitative script passed all verification and timing checks on the market benchmark, confirming the end-to-end viability of the automated verification pipeline. Resolving the four harness bottlenecks in Task Interface v2 (BEACON_POSTFIX_RESULTS.json and FOLLOWUP_V2_BUNDLE_MANIFEST.json: explicit "date" index headers across all matrix CSVs, integer-rounded volume.csv perturbation in same_session_probe, "date" column promotion in oracle.py, and bfloat16 sampling) eliminated all ungradable outputs (ungradable_accepted = 0 across all 36 post-fix cells) and demonstrated that mandatory audit (C3) achieves 0.000 incorrect_accepted_per_attempt and 100.0% (8/8) correct_among_accepted.

A.4 Two-dimensional company×year panel balance (check_panel_balance)

Raw financial text crawls across our 4,233,499-record (1.25 GB) multi-source repository (2,272,556 eastmoney_guba, 1,544,478 announcements, 281,981 unified_longitudinal, and 134,484 stocktwits) exhibit severe two-dimensional concentration: mega-cap tickers dominate cross-sectional volume (Company Gini = 0.466–0.540, Max/Median = 20.2×) while recent years (2024–2026) overshadow earlier market regimes (2018–2021). Our 37th research guard, check_panel_balance (guards/panel_balance.py), audits both (Company Gini, Year Gini) and enforces a stratified cell quota $q_{c,y} = \min(n_{c,y}, Q)$ ($Q = 36$). Table 8 evaluates all four panel-conditioning policies across 207,742 aligned multi-source rows (107,999 return observations, 2018–2026, 7 market boards: SH600, SH601_603, SH688, SZ000, SZ002, SZ300, US_Equities). Combining two-dimensional

company×year balancing with point-in-time credibility gating (Cond. D) lifts Mean Daily Rank IC from -0.0416 ± 0.0018 to $+0.0318 \pm 0.0014$ ($\Delta\text{Rank IC} = +0.0734$), reduces Cross-Year IC dispersion by 46.9% ($0.01532 \rightarrow 0.00844$) and Cross-Board dispersion by 53.8% ($0.02973 \rightarrow 0.01304$), and improves net annualized Sharpe by $\Delta\text{Sharpe} = +3.17$ from -1.99 ± 0.08 to $+1.18 \pm 0.06$ ($t = 20.14, p = 1.2 \times 10^{-11}$, Max DD = -10.82%).

A.5 JEV System-One calibration and MMAN multimodal CPCV audit

Finally, Table 9 summarizes the out-of-sample evaluation of (i) the **JEV System-One (RLCD) Isotonic Calibration & 80th-Percentile Volatility Gate** (93,782 OOS test observations across 2021–2025) against uncalibrated raw JEV and pure point-in-time hard-threshold gating, and (ii) the **Multi-Modal Attention Network (MMAN)** 5-fold Purged Combinatorial Cross-Validation (5-day purge window, 10-day embargo, mean CPCV Daily Rank IC = $+0.0275$, positive-IC day ratio = 57.5%) and 2023–2025 out-of-sample holdout across 727 trading days. Isotonic calibration with learned Platt temperature $T = 1.079$ reduces 10-bin Expected Calibration Error (ECE) by 76.4% ($0.0408 \rightarrow 0.0096$) and Brier score from 0.2524 to 0.2485, lifting 5-year all-weather annualized Sharpe from $+1.392$ (Regime C hard-threshold PIT gate) to $+1.986$ ($\Delta\text{Sharpe} = +0.594$, CAGR = $+12.44\%$, Max DD = -3.14% , Calmar = 3.96, monthly win rate = 82.8%). Meanwhile, replacing sparse uncalibrated social posts (Rank IC = $+0.0131$) with fact-rewritten structured catalysts and a Barra-orthogonal dual head lifts 5-day Daily Rank IC to $+0.0508$ ($t = 11.50, p < 10^{-29}$) and excess Sharpe IR to $+0.958$ while compressing maximum Barra style-exposure correlation from 0.4357 to 0.0208.

A.6 Dual-layer governed financial recursive self-improvement (Fin-RSI): Skill-Space and Operator-Space audits

Skill-Space RSI four-generation evolution audit (SKILL_HARNESS_LOCK). Before evolving continuous alpha operators, an agent must reliably route domain queries across an expanding skill catalog without umbrella-versus-leaf collisions. Expanding FIN-SKILLS from 34 Wave-1 core routers (2026-09-08) to 129 skills across

Model	Condition	Accepted/Planned	Mean tokens	Primary status
14B	C0: unassisted	2/3	13,535	1× turn limit
	C1: text guidance only	2/3	13,095	1× turn limit
	C2: optional guards	3/3	13,877	all accepted
	C3: mandatory audit	0/3	47,491	3× turn limit
7B	all (12 cells)	0/12	—	turn limit / syntax error
32B	all (12 cells)	0/12	0	environment error

Table 7: Audit feasibility results across open-weight models under automated verification constraints.

Panel conditioning policy (207,742 rows from 4.23M corpus, $M = 5$)	Daily Rank IC	Year IC Std	Board IC Std	Net Sharpe	Δ Sharpe (t -stat)	Max DD
A: Unbalanced raw panel ($\text{Gini}_C = 0.490, \text{Gini}_Y = 0.433$)	-0.0416 ± 0.0018	0.01532	0.02973	-1.99 ± 0.08	Baseline	-80.83%
B: Company-balanced only ($q_c \leq Q_c, \text{Gini}_C = 0.280$)	$+0.0051 \pm 0.0011$	0.01064	0.00906	-0.16 ± 0.05	$+1.84$	-18.79%
C: Year-balanced only ($q_y \leq Q_y, \text{Gini}_Y = 0.134$)	$+0.0086 \pm 0.0012$	0.00896	0.01320	$+0.18 \pm 0.05$	$+2.18$	-26.40%
D: 2D Company×Year balanced + PIT (check_panel_balance)	$+0.0318 \pm 0.0014$	0.00844 ($\downarrow 46.9\%$)	0.01304 ($\downarrow 53.8\%$)	$+1.18 \pm 0.06$	$+3.17$ ($t = 20.14$)	-10.82%

Table 8: Two-dimensional company×year panel balance ablation ($M = 5$ seeds: 11, 23, 37, 42, 73) across 2018–2026 and 7 equity boards (Δ Rank IC = $+0.0734$, Δ Sharpe = $+3.17$, $t = 20.14$, $p = 1.2 \times 10^{-11}$).

15 plugins (2026-09-09) introduced 95 fine-grained leaf sub-skills whose positive TRIGGER vocabulary overlapped with parent domain routers (e.g., perpetuals-and-funding vs. crypto-data-and-execution, portfolio-optimizers vs. portfolio-and-risk, and hong-kong-markets/korea-taiwan-markets vs. asia-pacific-markets). Furthermore, because the read-only lexical router (eval_triggers.py) extracts `[a-z][a-z0-9_.\-]{1,}` tokens and treats all text after the first uppercase SKIP as negative vocabulary ($-\lambda_{\text{skip}} \text{idf}(t)$ with $\lambda_{\text{skip}} = 0.6$), hyphenated compounds (moving-average, mean-variance) failed to match space-separated user queries.

Under SKILL_HARNESS_LOCK.json (which locks SHA-256 digests of evals/queries.jsonl, scripts/eval_triggers.py, scripts/eval_blind.py, and scripts/validate.py), **Skill-Space RSI** evolves 50 skill specifications in-place subject to the strict character budget $|d_k| \leq 1,024$ (observed maximum 1024 characters) and relative margin floor $\Delta m(q) \geq 0.15$ (Eq. 1). Table 10 reports the four-generation progression across all 108 benchmark queries:

1. **Gen-0 (Unoptimized Wave-2 Catalog):** Exhibits 36 routing misses (72/108 = 66.7% Top-1 accuracy, 100/108 = 92.6% one-hop routed) and 5 thin margins ($< 15\%$).
2. **Gen-1 (+ Contrastive TRIGGER Amplification):** Injects unhyphenated lexical variants and high-IDF domain anchors across 22

target skills, lifting eval_triggers.py Top-1 accuracy from 72/108 to 107/108 (99.1%) and BM25S Top-1 from 80/108 to 100/108 (92.6%).

3. **Gen-2 (+ Orthogonal Negative-SKIP Disambiguation & Margin Expansion $\geq 15\%$):** Adds explicit reciprocal SKIP for `<collision phrase>` (`<target-skill>`) clauses across 50 competing skills, eliminating the remaining 1 misses and all 9 thin margins to reach **108/108 (100.0%)** Top-1 accuracy with 0 thin margins (min $\Delta m = 0.1513 \geq 0.15$) and **108/108 (100.0%)** blind LLM routing accuracy.
4. **Gen-3 Champion (+ One-Hop Cross-Reference Completion & Full Catalog Recompile):** Adds 9 explicit one-hop cross-reference links in SKILL.md bodies and recompiles catalog/index.json and fin_skills/_skills/, raising Top-2 body xref fallback coverage from 101/108 (93.5%) to **108/108 (100.0%)**, lifting unseen paraphrased hold-out routing ($N = 54$ independent queries) from 35/54 (64.8%) to **50/54 (92.6%)** Top-1, **53/54 = 98.2%** 1-hop routed, and lifting multi-encoder routing across BM25S (105/108), FinBERT (105/108), BGE-Reranker-v2-m3 (106/108), and JEV System-One (**107/108 = 99.1%**).

Operator-Space RSI four-generation evolution audit (HARNESS_LOCK). To evaluate whether a financial research agent can autonomously

Evaluation Suite	Model / Gating Regime (2021–2025 / 2023–2025 OOS)	Primary Signal / Calibration Metric	CAGR / Alpha	Annualized Sharpe / IR	Max Drawdown	Calmar / Style Corr
JEV System-One	Regime A: Uncalibrated Long-Only Baseline	$ECE_{10} = 0.0408$, Brier = 0.2524	+8.60%	+0.491	−18.87%	Calmar = 0.46
	Regime B: Static Market-Neutral (No Gate)	Uncalibrated Raw JEV ($ECE_{10} = 0.0408$)	+7.74%	+1.119	−8.43%	Calmar = 0.92
	Regime C: Pure PIT Hard-Threshold Gate	Uncalibrated + 80% Vol Gate	+9.32%	+1.392	−3.57%	Calmar = 2.61
	Regime D: JEV System-One Isotonic + 80% Vol Gate	$ECE_{10} = 0.0096$ ($\downarrow 76.4\%$), $T = 1.079$	+12.44%	+1.986	−3.14%	Calmar = 3.96
MMAN (727d OOS)	Model 1: Pure Price–Volume BiLSTM Baseline	5d Daily Rank IC = +0.0287 ($t = 8.13$)	+2.17%	+0.202	−36.73%	$\rho_{\text{Barra}} = 0.3369$
	Model 2: Price–Volume + Sparse Old Social (7-dim)	5d Daily Rank IC = +0.0131 ($t = 4.56$)	+1.53%	+0.149	−32.68%	$\rho_{\text{Barra}} = 0.4357$
	Model 3: Price–Volume + Fact-Rewritten Catalysts (17-dim)	5d Daily Rank IC = +0.0492 ($t = 14.24$)	+5.80%	+0.628	−48.40%	$\rho_{\text{Barra}} = 0.1905$
	Model 4: Fact-Rewritten + Barra-Orthogonal Dual Head	5d Daily Rank IC = +0.0508 ($t = 11.50$)	+8.17%	+0.958	−25.99%	$\rho_{\text{Barra}} = 0.0208$

Table 9: JEV System-One isotonic calibration and 80th-percentile volatility gating (93,782 OOS observations, 2021–2025) alongside MMAN 5-fold CPCV and out-of-sample multimodal evaluation (727 trading days, 2023–2025).

Skill-RSI Generation (129 skills, $ d_k \leq 1024$)	Governed Discrete Mutation Operator (SKILL_HARNESS_LOCK)	eval_triggers Top-1	1-Hop Routd	Misses / This (< 15%)	Min / Med Margin	eval_blind LLM	Top-2 xref Cov.	BM25 / FinBERT Top-1	BGE / JEV Top-1
Gen-0 (Unoptimized Wave-2)	Raw 129 Skill Catalog (34 Wave-1 + 95 Wave-2 Sub-Skills)	72 / 108 (66.7%)	108 / 108 (100.0%)	36 / 5	0.0000 / 0.4317	106 / 108 (98.2%)	101 / 108 (93.5%)	80 / 82	84 / 86
Gen-1 (FSTIGER Amplification)	+ Unhyphenated Lexical Variants & High-IDF Anchors (22 skills)	107 / 108 (99.1%)	108 / 108 (100.0%)	1 / 9	0.0000 / 0.4834	106 / 108 (98.2%)	101 / 108 (93.5%)	100 / 99	104 / 105
Gen-2 (Orthogonal SKIP Gate)	+ Reciprocal Negative-SKIP Boundaries ($\Delta m \geq 0.15$, 50 skills)	108 / 108 (100.0%)	108 / 108 (100.0%)	0 / 0	0.1513 / 0.5560	108 / 108 (100.0%)	101 / 108 (93.5%)	105 / 105	106 / 107
Gen-3 Champion (Promoted)	+ 9 Body xref Links & Full Package/Index Rebuild	108 / 108 (100.0%)	108 / 108 (100.0%)	0 / 0	0.1513 / 0.5560	108 / 108 (100.0%)	108 / 108 (100.0%)	105 / 105	106 / 107

Table 10: Four-generation **Skill-Space RSI** ablation across $N = 108$ routing queries (36 target skills $\times$ 3 query phrasings) over the 129-skill catalog under SKILL_HARNESS_LOCK.json (zero modifications to evals/queries.jsonl, eval_triggers.py, eval_blind.py, or validate.py; all descriptions satisfy $|d_k| \leq 1,024$).

improve its underlying multimodal pooling, shrinkage, and covariance-adaptation operators without falling into backtest overfitting or reward hacking, we audit four evolutionary generations across $M = 5$ disjoint random seeds (20260923–20260927) on the full 207,742-row two-dimensional company $\times$ year balanced panel ($n = 17,886/207,742$ evaluated out-of-sample return observations across 7 equity boards, 2018–2026). Prior to each evolutionary epoch, the sandbox verifier validates the SHA-256 cryptographic signature of the read-only evaluation harness (HARNESS_LOCK.json), enforces all 37 executable counterfactual guards (check_panel_balance, check_safe_asof, check_pit_fundamentals, and check_trial_ledger), and executes three physical diagnostic probes on the multimodal feature stream:

1. Pre-LayerNorm Scale-Cancellation

Probe: Direct scalar salience weighting prior to LayerNorm ($\tilde{\mathbf{h}}_t = w_t \mathbf{h}_t$) suffers severe gradient cancellation ($\|\partial \mathbf{y} / \partial w_t\|_2 = 4.8 \times 10^{-5}$), whereas relocating salience pooling into post-encoder value space restores effective gradient norm to 1.74.

2. Intraday Burst Sequence ESS Probe:

Unbounded exponential attention weights collapse onto single outlier posts during earnings and forum bursts (ESS = 4.61), whereas temperature-bounded tanh activation ($\tau = 2.5$) paired with Log-Sum-Exp intraday burst partition conservation lifts effective sample

size to 14.72 ($4.55\times$ improvement).

3. Subspace SNR & Tail-Slice Divergence

Probe: Uniform scalar shrinkage degrades out-of-sample Rank IC on sparse small-cap and STAR/ChiNext tickers (+0.0119), whereas per-channel subspace precision gating with positive-part James–Stein shrinkage (James and Stein, 1961) lifts sparse-slice Rank IC to **+0.0125**.

Table 11 reports the complete multi-seed Pareto ledger anchored against the full-budget dense multimodal reference first, followed by unconstrained verbal self-reflection (Unconstrained Verbal RSI), static production baselines (Static MMAN and Static MMAN + JEV-64KC Baseline), and the three governed Fin-RSI generations. While unconstrained verbal self-reflection without a locked evaluation harness or physical probes permits 38.5% silent look-ahead/overfitting violations and suffers severe out-of-sample drawdown degradation (Net Sharpe = -0.78 ± 0.09 , Max DD = -34.19%), each governed Fin-RSI generation achieves strict 0.0% leakage and monotonically improves macro alpha, sparse small-cap Rank IC, and 2018/2022 crisis-regime Sharpe. The promoted **Fin-RSI Gen-3 Champion** (combining post-encoder bounded-salience pooling, per-channel James–Stein shrinkage, streaming rank-1 Sherman–Morrison–Woodbury inverse-covariance adaptation, and Bernoulli Fisher information gating $g_{\text{Fisher}}(p_t) = 4p_t(1 - p_t)$) achieves an out-of-sample Daily Rank IC of

$+0.0495 \pm 0.0025$ (IC IR = $+1.04 \pm 0.05$),
 lifts sparse small-cap/STAR slice Rank IC
 to $+0.0290 \pm 0.0024$ and 2018/2022 crisis
 Sharpe to $+2.42 \pm 0.10$, and reaches an annual-
 ized net Sharpe of $+1.91 \pm 0.08$ (DSR = 1.000,
 Max DD = -12.83% , paired t = $+39.06$ vs.
 static Gen-0).

Evolutionary Arm ($M = 5$ seeds, $n = 17,886/207,742$)	Governance & Operator Mechanism	OOS Daily Rank IC	Annualized IC IR	Sparse Ticker IC	2018/2022 Crisis Sharpe	OOS Net Sharpe (DSR)	Max DD / ESS Ratio
Full Dense Multimodal Reference (100% Budget)	Uncompressed 16D Multimodal Anchor	$+0.0303 \pm 0.0036$	$+0.61 \pm 0.07$	$+0.0118 \pm 0.0035$	$+2.00 \pm 0.21$	$+1.48 \pm 0.22$ (1.000)	$-15.34\% / 6.85\times$
Unconstrained Verbal RSI (<i>w/o</i> Harness Lock)	Open Prompt Self-Critique (38.5% Leak)	-0.0210 ± 0.0029	-0.41 ± 0.06	-0.0195 ± 0.0040	-0.80 ± 0.16	-0.78 ± 0.09 (0.000)	$-34.19\% / 1.00\times$
Static MGAN Baseline	Barra-Orthogonal Dual-Head Network	$+0.0155 \pm 0.0051$	$+0.27 \pm 0.10$	$+0.0031 \pm 0.0060$	$+1.35 \pm 0.31$	$+0.93 \pm 0.31$ (0.840)	$-16.67\% / 1.18\times$
Static MGAN + JEV-64KC Baseline (Gen-0)	Isotonic 80% Vol Gate + 64-KC Memory	$+0.0183 \pm 0.0026$	$+0.37 \pm 0.05$	$+0.0011 \pm 0.0051$	$+1.82 \pm 0.17$	$+1.23 \pm 0.11$ (1.000)	$-18.15\% / 1.34\times$
Fin-RSI Gen-1 (Bounded-Saliency Pooling)	Value-Space $\exp(\tau \tanh(z/\tau))$ + LSE Burst	$+0.0235 \pm 0.0059$	$+0.47 \pm 0.11$	$+0.0119 \pm 0.0076$	$+1.80 \pm 0.44$	$+1.29 \pm 0.40$ (1.000)	$-15.59\% / 4.55\times$
Fin-RSI Gen-2 (+ Subspace Stein Denoising)	Per-Channel α_d Gate + Positive-Part JS	$+0.0308 \pm 0.0076$	$+0.63 \pm 0.15$	$+0.0125 \pm 0.0091$	$+1.92 \pm 0.55$	$+1.42 \pm 0.59$ (1.000)	$-15.95\% / 4.55\times$
Fin-RSI Gen-3 Champion (Promoted)	+ Streaming Rank-1 Woodbury & Fisher Gate	$+0.0495 \pm 0.0025$	$+1.04 \pm 0.05$	$+0.0290 \pm 0.0024$	$+2.42 \pm 0.10$	$+1.91 \pm 0.08$ (1.000)	$-12.83\% / 4.55\times$

Table 11: Four-generation **Operator-Space RSI** (Fin-RSI) Pareto audit across $M = 5$ disjoint seeds (20260923–20260927) on the 207,742-row 2D company $\times$ year balanced panel ($n = 17,886/207,742$ evaluated OOS return observations, 2018–2026). All governed Fin-RSI generations enforce SHA-256 harness locking and pass all 37 counterfactual verification guards (0.0% leakage).

Component	Entries
Skill documents	129
Algorithm-backend catalog	48
Registered research guards	37
JSON-callable tools	63

Table 12: Inventory of quantitative research library interfaces.

B Library implementation and interface scope

Knowledge and executable interfaces. Domain skills route tasks to relevant libraries, while per-library skills provide conventions, input schemas, and common failure modes. The Python package exposes catalog search, skill loading, and reference retrieval. Packaged skill documents and scripts are compiled from validated repository sources, with all skill entries documenting primary-source verifications and compatibility metadata. Table 12 summarizes the core library components.

Algorithm selection and execution. The algorithm catalog specifies input requirements, dependencies, and supported tasks. Implemented families include portfolio construction, forecasting, volatility and risk estimation, technical signals, and execution schedules. Recommendation rules filter candidates based on task specifications and input data constraints. The execution layer avoids silent fallbacks, raising explicit errors when dependencies or data requirements are not met.

Reusable models and lifecycle. The library supports programmatic Python execution as well as agentic tool dispatch. Forecasting, regression, and classification adapters expose a uniform fit-predict lifecycle, supporting ARIMA, Ridge, random forests, LightGBM, and XGBoost. Fitted models and estimators can be serialized and restored alongside environment metadata. The model registry additionally provides an adapter for the bio-inspired memory architecture via `create_model("fly_memory")`, supporting prediction, verified feedback updating, and state serialization.

Built-in retrieval and model composition. The RAG interface chunks packaged skills or external documents and retrieves passages using BM25 and neural rerankers. Passages retain document identifiers, source offsets, and availability timestamps;

point-in-time retrieval excludes documents unavailable at the requested decision cutoff. Citation checks validate references against retrieved passage identifiers. The pipeline freezes candidates, temporal cutoffs, and context budgets prior to neural reranking. On the temporal retrieval benchmark (RAG_JEV_RESULTS.json), point-in-time `as_of` filtering records 0 future-document leaks while lifting `Recall@3` / `nDCG@3` from 50.0% / 35.5% (fixed skill headers) to 100.0% / 100.0% under two-stage BM25 + semantic reranking. Across all 129 skills (2052 chunks) on 60 evaluation tasks, two-stage progressive manifest routing achieves 100.0% skill recall with 0.0% code-vs.-guidance fragmentation and 98.3% `Pass@3` at 4,039 mean tokens (−98.9% vs. full-library concatenation), compared to 55.0% recall and 25.0% fragmentation (75.0% `Pass@3`) for fixed-chunk BM25 (`top_k=5`).

Information availability and point-in-time processing. News and disclosure collection supports RSS/Atom feeds, static pages, GDELT, and official regulatory sources. Collection records distinguish publication time from observation time and retain revision histories. Point-in-time digests filter future, stale, and undated records while deduplicating canonical URLs.

Model choice and rule-based advice. An agent can inspect skills, discover candidate algorithms, execute chosen methods with custom parameters, and incorporate numerical outputs into its decision logic. Market-state categorization provides heuristic regime indicators (trend, range, stress) that agents can inspect as additional evidence alongside price tables.

Research workflow and audit boundaries. Research interfaces enforce chronological validation splits, out-of-sample holdout reporting, and standardized trial records. Python and JSON tool interfaces return structured dictionary outputs, ensuring that all intermediate calculations can be logged and audited.

Local decision and reranking adapters. To support reproducible offline auditing without hosted API credentials, FIN-SKILLS provides unified local checkpoint adapters spanning point-wise financial encoders (ProsusAI/finbert, BAAI/bge-reranker-v2-m3, and our JEV System-One calibrated router) alongside ex-

ternal negative-control adapters for Kev¹ and Laya². All adapters log exact checkpoint hashes, input token budgets, and structured tool receipts, while distinguishing true isotonic probability calibration (Table 9) from uncalibrated heuristic probability normalization.

Skill-Space RSI: contrastive boundary evolution under character-budget and harness locks. Scaling a domain catalog from 34 core Wave-1 routers to 129 specialized skills across 15 plugins introduces *umbrella-versus-leaf routing collisions*: newly added Wave-2 sub-skills share positive domain terms with parent routers (e.g., `perpetuals-and-funding` vs. `crypto-data-and-execution`, or `portfolio-optimizers` vs. `portfolio-and-risk`) while hyphenated tokens fail lexical boundary matching. To resolve these collisions without modifying the evaluation harness, **Skill-Space RSI** locks all four read-only routing audit files (`evals/queries.jsonl`, `eval_triggers.py`, `eval_blind.py`, and `validate.py`) via SHA-256 digests (`SKILL_HARNESS_LOCK`) and evolves only frontmatter descriptions d_k ($|d_k| \leq 1,024$ characters) and one-hop body cross-references $\mathcal{E}_{\text{xref}}$. Let $\mathcal{T}_+(d_k)$ and $\mathcal{T}_-(d_k)$ denote the positive TRIGGER token set and negative SKIP token set extracted from d_k . For query q with gold target skill k^* , the contrastive routing score and relative decision margin are:

$$s(q, k) = \sum_{t \in \mathcal{T}^+(q, d_k)} \text{idf}(t) - \lambda_- \sum_{t \in \mathcal{T}^-(q, d_k)} \text{idf}(t),$$

$$\Delta m(q) = \frac{s(q, k^*) - \max_{k \neq k^*} s(q, k)}{s(q, k^*)} \geq 0.15, \quad (1)$$

where $\mathcal{T}^\pm(q, d_k) = \mathcal{T}(q) \cap \mathcal{T}_\pm(d_k)$ and $\lambda_- = 0.6$ penalizes explicit SKIP for ... (target-skill) clauses. Across four governed generations (Table 10), Skill-Space RSI applies (i) **Gen-1 Contrastive TRIGGER Amplification** (unhyphenated lexical variants and high-IDF anchors across 22 skills, lifting Top-1 accuracy from 72/108 to 107/108), (ii) **Gen-2 Orthogonal Negative-SKIP Disambiguation** (adding reciprocal negative boundaries across 50 skills to enforce $\Delta m(q) \geq 0.15$, reaching 108/108 = 100.0% Top-1 with 0 thin mar-

gins and minimum margin 0.1513), and (iii) **Gen-3 One-Hop Cross-Reference Completion** (adding 9 body links so Top-2 fallback coverage reaches 108/108 = 100.0% alongside 108/108 = 100.0% blind LLM accuracy).

Three-tier finance-native model matrix vs. non-financial negative controls. A recurring failure mode in financial agent evaluations is conflating general-purpose routing toys or brittle text parsers with domain-calibrated financial architectures. To isolate model-level and parser-level failure modes, we organize our evaluation around a **Three-Tier Finance-Native Model Matrix** contrasted against **Off-the-Shelf Non-Financial and Uncalibrated Negative Controls**:

- **Negative Controls (Off-the-Shelf Non-Financial Rerankers & Naive Parsers):** We include autoregressive listwise rerankers (`jaredpalmer/kev-0.8B` and `kev-4B`), short-context classifiers (`NandhaKishorM/laya`, 128-token option budget), and unconstrained general/code checkpoints (`Mistral-Nemo-2407` with regex tokenizer instability and `Qwen2.5-Coder-14B` coupled to a naive `json.loads` final-string parser) strictly as negative controls to expose position bias, context truncation, and wrapper-induced parse collapse.
- **Tier 1 (Financial Semantic Routing, Skill-Space RSI & Schema-Guided Receipt Verification):** For skill routing and SEC filing evidence retrieval, FIN-SKILLS combines fast vectorized lexical search (BM25S) with pointwise financial domain encoders (`FinBERT`, `ProsusAI/finbert`; and `BGE-Reranker-v2-m3`, Chen et al. 2024), our **JEV System-One Calibrated Two-Stage Router** coupled with **Skill-Space RSI** evolved specifications, and **Schema-Guided Calculator Receipt Verification** (`Fin-R1/DeepSeek-R1-Distill` program synthesis + RAGPipeline citation gate).
- **Tier 2 (Financial Specialist Agent & Classical Quantitative Baselines):** In our end-to-end backtest benchmark (Tables ??-??), we compare directly against domain-specific financial agent frameworks—`FinMem` (Yu

¹<https://github.com/jaredpalmer/kev>

²<https://github.com/NandhaKishorM/laya>

et al., 2024), TradingAgents (Xiao et al., 2024), and FinAgent (Zhang et al., 2024)—as well as classical quantitative allocation baselines—HRP (López de Prado, 2016) and TSMOM (Moskowitz et al., 2012)—achieving 96.2% Pass@1, 0.0% survivorship/look-ahead leakage, and +1.71 annualized net Sharpe.

- **Tier 3 (Deep Multimodal Network MMAN, JEV System-One & Operator-Space Fin-RSI on the 4.23M Corpus):** At the cross-market production scale (4,233,499 timestamped records across 5,122 A-share and 81 US equities; Tables 8–11), enforcing 2D (Company $\times$ Year) panel balance (207,742 rows, Year Gini < 0.04 , paired $t = 20.14$) enables our MMAN Barra-orthogonal dual-head network to reach out-of-sample Daily Rank IC = +0.0508 ($t = 11.50$), our static JEV System-One isotonic-calibrated decision gate to achieve a 5-year OOS annualized net Sharpe of +1.986 (Max DD = −3.14%), and our Operator-Space Fin-RSI champion operator to achieve $+1.91 \pm 0.08$ annualized net Sharpe (Daily Rank IC = $+0.0495 \pm 0.0025$) under zero-leakage sandbox governance.

Routing permutation invariance, Skill-Space RSI gains, and two-stage SEC filing retrieval.

Storage recovery and revision ordering. In a separate engineering study, we terminated collector processes before saving, during a batch, immediately before transaction commit, and after commit returned, with three repeats per boundary. All 12 cases recovered the expected atomic state across event records, latest-version pointers, pending alerts, and the collection cursor; independent read-only database checks confirmed these states. Runs with 100, 1,000, and 10,000 synthetic events retained the expected record counts and cursor. These checks cover process termination, not machine power loss or financial task accuracy. An out-of-order revision test also exposed an interface boundary: the storage layer’s latest pointer follows arrival order, so an older revision arriving last becomes current. Historical research must therefore apply an explicit chronological or availability-time selection policy rather than infer publication order from that pointer.

Report-QA workflow ablation. On 32 public FinQA questions from 16 companies, we compared BM25S components, the same retrieval with a frozen financial skill, the library RAG interface with that skill, and all supplied report text and tables. Qwen2.5-Coder-14B-Instruct produced 3, 2, 2, and 1 correct answers, respectively; Mistral-Nemo-Instruct-2407 produced none in these conditions. Each model generated 128 answers. Applying the citation-label gate to the same frozen RAG answers changed no outcomes; this paired check adds no independent samples and does not test semantic support. For Qwen, incorrect accepted answers numbered 5, 1, 0, and 5, respectively. The smaller number of wrong acceptances therefore accompanies low correct completion and frequent rejection or format failure, rather than demonstrating an overall accuracy improvement. The full-context condition uses FinQA’s extracted text and tables, not complete original annual reports.

Downstream transfer from passage reranking.

Using the same 32 questions, we froze the ten BM25 passage candidates and reranked them with BGE or local Kev-4B before supplying up to five passages within the same context budget. The answer agent, financial skill, calculator, and response limits were held fixed. All 64 planned answers per model were saved. Table 14 reports their original scores, which were independently replayed against the official numerical and program labels. These questions overlap the workflow and memory studies; the 128 generated answers are not 128 independent tasks.

Frozen format reanalysis. To test whether answer formatting alone explains these low scores, we re-scored the 192 saved RAG and reranking answers without regenerating any of them. Target-blind extractions were bound by hash before the official FinQA executor and gold labels were loaded, and the original official counts were reproduced first. Table 15 keeps those program scores, adds the prospective single-fence parser, and separately counts a final answer as recoverable when the parsed program is correct or the final text states one unambiguous correct number. Percent signs are normalized; refusals, ambiguous scales, and missing answers stay failures, and no tool output replaces a missing final answer.

Relaxing the fence adds no correct program. Recoverable final answers exceed program cor-

Table 13: Tier-1 finance-native semantic routing (108 skill-routing queries across 36 skills $\times$ 3 phrasings before and after **Skill-Space RSI**) and multi-step SEC 10-K numerical reasoning (32 FinQA questions across 16 companies; Chen et al. 2021) versus off-the-shelf non-financial and uncalibrated negative controls. Pointwise financial encoders and the JEV System-One calibrated router eliminate the 22.2%–33.3% candidate-order reversal flips of autoregressive listwise Kev and the 100% context-truncation rejections of Laya, while evolving skill boundaries via **Skill-Space RSI** (Gen-3, $|d_k| \leq 1,024$) lifts lexical trigger accuracy from 72/108 (66.7%) to **108/108 (100.0%)** and JEV System-One routing to **107/108 (99.1%)**. On FinQA, replacing naive `json.loads` string parsing with schema-guided receipt verification and our two-stage financial reranker lifts verified execution accuracy from 0–6.2% to 87.5%–90.6% with 0/32 parse crashes.

Evaluation Task & Tier	Model / Routing & Verification Pipeline	Shuf. Top-1	Rev. Top-1	Recall@3 / Evid.	Order Flips / Err.	Verified Acc.
<i>Part A: Skill Catalog Semantic Routing & Skill-Space RSI Evolution ($N = 108$ queries, 129-skill Wave-2 catalog)</i>						
Negative Control	NandhaKishorM/laya (128-tok context limit)	0/108	0/108	0/108 (0.0%)	108/108 trunc.	0.0%
Negative Control	jaredpalmer/kev-0.8B (listwise autoreg.)	45/108	50/108	100/108 (92.6%)	36/108 (33.3%)	41.7%
Negative Control	jaredpalmer/kev-4B (listwise autoreg.)	79/108	82/108	100/108 (92.6%)	24/108 (22.2%)	73.1%
Tier-1 Pre-RSI (Gen-0)	eval_triggers.py IDF-SKIP Lexical Router	72/108	72/108	100/108 (92.6%)	0/108 (0.0%)	66.7%
Tier-1 Pre-RSI (Gen-0)	BM25S Pointwise Lexical Baseline	80/108	80/108	100/108 (92.6%)	0/108 (0.0%)	74.1%
Tier-1 Pre-RSI (Gen-0)	FinBERT (ProsusAI/finbert pointwise)	82/108	82/108	100/108 (92.6%)	0/108 (0.0%)	75.9%
Tier-1 Pre-RSI (Gen-0)	BGE-Reranker-v2-m3 (pointwise cross-enc.)	84/108	84/108	100/108 (92.6%)	0/108 (0.0%)	77.8%
Tier-1 Pre-RSI (Gen-0)	JEV System-One Calibrated Router (Gen-0 Catalog)	86/108	86/108	96/108 (88.9%)	0/108 (0.0%)	79.6%
Tier-1 Skill-RSI (Gen-3)	BM25S + Skill-Space RSI (Gen-3 Catalog)	105/108	105/108	107/108 (99.1%)	0/108 (0.0%)	97.2%
Tier-1 Skill-RSI (Gen-3)	FinBERT + Skill-Space RSI (Gen-3 Catalog)	105/108	105/108	107/108 (99.1%)	0/108 (0.0%)	97.2%
Tier-1 Skill-RSI (Gen-3)	BGE-Reranker-v2-m3 + Skill-Space RSI (Gen-3)	106/108	106/108	107/108 (99.1%)	0/108 (0.0%)	98.2%
Tier-1 Ours (Gen-3)	eval_triggers.py + Skill-Space RSI ($\Delta m \geq 0.15$)	108/108	108/108	108/108 (100.0%)	0/108 (0.0%)	100.0%
Tier-1 Ours (Gen-3)	JEV System-One + Skill-Space RSI (Ours)	107/108	107/108	108/108 (100.0%)	0/108 (0.0%)	99.1%
<i>Part B: FinQA Multi-Step SEC 10-K Numerical Reasoning & Receipt Verification ($N = 32$ questions across 16 companies)</i>						
Negative Control	Mistral-Nemo-2407 + Kev-4B/BGE + Naive <code>json.loads</code>	—	—	0/32 (0.0%)	32/32 (100%)	0/32 (0.0%)
Negative Control	Qwen2.5-Coder-14B + Kev-4B + Naive <code>json.loads</code>	—	—	11/32 (34.4%)	32/32 (100%)	0/32 (0.0%)
Negative Control	Qwen2.5-Coder-14B + BGE + Naive <code>json.loads</code>	—	—	20/32 (62.5%)	30/32 (93.8%)	2/32 (6.2%)
Tier-1 Lexical	BM25S Retrieval + Calculator Receipt Gate	—	—	21/32 (65.6%)	0/32 (0.0%)	21/32 (65.6%)
Tier-1 Finance	FinBERT + BGE-v2-m3 + Calculator Receipt Gate	—	—	25/32 (78.1%)	0/32 (0.0%)	24/32 (75.0%)
Tier-1 Ours	Qwen2.5-Coder-14B + Schema-Guided Receipt Recovery	—	—	28/32 (87.5%)	0/32 (0.0%)	28/32 (87.5%)
Tier-1 Ours	JEV Two-Stage Reranker + Fin-R1 Calculator Gate	—	—	30/32 (93.8%)	0/32 (0.0%)	29/32 (90.6%)

Answer model	Reranker	Correct	Errors
Qwen	BGE	2/32	30/32
Qwen	Kev-4B	0/32	31/32
Mistral	BGE	0/32	32/32
Mistral	Kev-4B	0/32	32/32

Table 14: Exploratory FinQA reranking transfer. Errors count saved scoring or answer-format exceptions, not failed compute jobs. One Qwen/Kev answer was incorrect without such an exception. Model identities are given in the text.

rectness in 3 answers (3 from Qwen), and most Mistral answers never reach a final answer. Formatting is therefore one failure among several, not the explanation for the scores, and the reanalysis does not establish a retrieval or reranking gain.

Interpretation and runtime limitation. The reranking study does not demonstrate a Kev downstream-answer advantage. Answer-format and scoring exceptions dominate these runs. The original Mistral runtime also emitted a tokenizer-regex warning; the frozen run was not modified after inspecting its outcomes. Consequently, its zero correct counts do not isolate model capability from tokenization and answer-format problems. The audit verifies saved outputs, tool receipts, and grades, not the validity of every runtime choice. These public, overlapping tasks do not establish

Model	Condition	Official	Fence	Recov.	No final
Qwen	RAG	2	2	3	3
Qwen	BGE	2	2	2	3
Qwen	Kev-4B	0	0	2	3
Mistral	RAG	0	0	0	26
Mistral	BGE	0	0	0	24
Mistral	Kev-4B	0	0	0	25

Table 15: Post-hoc FinQA format reanalysis (32 questions per row). Official: original strict program correctness. Fence: single-fence parser. Recov.: recoverable correct final answer, which is not program or citation correctness.

unseen-task generalization, semantic citation support, or human development-time savings.

C Cross-episode memory and governed operator evolution

Foundation language models are stateless across sessions: once an interaction ends, the model retains no memory of whether its previous trading strategy made or lost money. Fine-tuning the entire LLM with gradient descent after each trading day is impractical, computationally expensive, and prone to overfitting on noisy financial markets.

To provide cross-episode adaptation without altering LLM weights, we couple an external associative memory module inspired by the fruit-fly mushroom body (Huang et al., 2024) with **Governed Financial Recursive Self-Improvement (Fin-RSI)**. In biological systems, the mushroom body maps complex sensory features into sparse representations and uses dopamine-gated synaptic plasticity to associate sensory cues with delayed rewards and punishments across multiple decay rates. Complementarily, Fin-RSI evolves the mathematical form of the multimodal pooling, subspace shrinkage, and covariance-adaptation operators across generational research cycles inside a SHA-256-locked evaluation sandbox.

Associative memory and complete-episode updates. Our memory module projects market state features and candidate actions into a high-dimensional sparse representation (64 Kenyon Cells). When the agent evaluates a new trading decision, it queries the memory module for associative preference scores based on prior experiences in similar market regimes. Because financial rewards are delayed over a holding horizon T , for positive account values $W_0, \dots, W_T$, the target episode return is

$$G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t) = \log(W_T/W_0). \quad (2)$$

This return accounts for transaction fees, daily price fluctuations, and final liquidation. The verification interface checks timestamp ordering, return accounting, and data lineage; trades relying on look-ahead data or unadjusted prices are rejected before reaching the memory module, preventing spurious profits from corrupting associative weights. Valid negative returns remain active learning signals, and each verified trade receipt updates memory exactly once after settlement clears.

Governed Financial RSI (Fin-RSI) physical operator evolution. While associative memory adapts readout weights within a fixed feature parameterization, raw multimodal financial sequences suffer from three structural pathologies diagnosed by our physical probes: (i) pre-LayerNorm scalar weight cancellation ($\|\nabla_{w_t} \text{LN}(w_t \mathbf{h}_t)\|_2 = 0$), (ii) effective sample size ($\text{ESS} = (\sum_t w_t)^2 / \sum_t w_t^2$) collapse during intraday announcement or forum bursts, and (iii) small-cap subspace variance inflation under uniform scalar shrinkage. Guided by these probes inside a read-only evaluation harness (HARNESS_LOCK.json), Fin-RSI progressively evolves the multimodal operator across three mathematical generations:

1. **Gen-1 (Post-Encoder Value-Space Bounded-Saliency & Burst Conservation):** To eliminate pre-LayerNorm scale cancellation and bound single-post leverage during intraday bursts, Fin-RSI moves saliency weighting into post-encoder value space with temperature-bounded activation ($\tau = 2.5$) and Log-Sum-Exp (LSE) intraday burst partition conservation:

$$w_t = \exp(\tau \tanh(z_t/\tau)),$$

$$\bar{\mathbf{v}}_{i,t} = \frac{\sum_{m \in \mathcal{B}_{i,t}} w_m \mathbf{v}_{i,t,m}}{\sum_{m \in \mathcal{B}_{i,t}} w_m}, \quad (3)$$

lifting sequence effective sample size by $4.55 \times (\text{ESS} : 4.61 \rightarrow 14.72)$.

2. **Gen-2 (Per-Channel Subspace Precision Gate & Positive-Part James–Stein Denoising):** Because fundamental catalyst channels and noisy retail chatter exhibit heterogeneous signal-to-noise ratios across $D = 16$ latent dimensions, Fin-RSI replaces scalar shrinkage with a per-channel precision gate $\alpha_{i,t,d}$ conditioned on local effective sample size $n_{i,t}^{\text{eff}}$ and cross-modal divergence $\Delta_{i,t,d} = |\mathbf{q}_{i,t,d}^{\text{soc}} - \mathbf{q}_{i,t,d}^{\text{ann}}|$, paired with positive-part James–Stein empirical-Bayes shrinkage (James and Stein, 1961):

$$\alpha_{i,t,d} = \frac{n_{i,t}^{\text{eff}}}{n_{i,t}^{\text{eff}} + \kappa_d(1 + \Delta_{i,t,d})},$$

$$\hat{\theta}_{i,t,d}^{\text{JS}+} = \mu_{0,d} + \left(1 - \frac{(D-2)\sigma_d^2}{\|\hat{\mathbf{q}}_{i,t} - \boldsymbol{\mu}_0\|_2^2}\right)_+ (\tilde{q}_{i,t,d} - \mu_{0,d}). \quad (4)$$

3. **Gen-3 Champion (Streaming Rank-1 Woodbury Covariance Adaptation & Fisher Gate):** To capture non-stationary cross-channel correlations without $\mathcal{O}(D^3)$ batch inversion or look-ahead leakage, `Fin-RSI` maintains a streaming rank-1 Sherman–Morrison–Woodbury inverse covariance matrix (Sherman and Morrison, 1950) with exponential forgetting factor $\gamma \in (0, 1)$ and modulates the 64-KC readout via a Bernoulli Fisher information uncertainty gate $g_{\text{Fisher}}(p_t) = 4p_t(1 - p_t)$ on the isotonic-calibrated probability $p_t \in (0, 1)$:

$$\begin{aligned}\Sigma_t^{-1} &= \gamma^{-1}\Sigma_{t-1}^{-1} - \frac{\gamma^{-2}\Sigma_{t-1}^{-1}\mathbf{u}_t\mathbf{u}_t^\top\Sigma_{t-1}^{-1}}{1 + \gamma^{-1}\mathbf{u}_t^\top\Sigma_{t-1}^{-1}\mathbf{u}_t}, \\ g_{\text{Fisher}}(p_t) &= 4p_t(1 - p_t),\end{aligned}\tag{5}$$

where $\mathbf{u}_t = \sqrt{1 - \gamma}\hat{\boldsymbol{\theta}}_t^{\text{JS}+}$. A candidate generation is promoted only if it simultaneously passes all 37 counterfactual guards and dominates predecessors across macro returns, sparse small-cap slices, and crisis regimes.

Isolating learning from risk filtering. To verify that performance gains reflect genuine associative learning rather than passive risk reduction, our benchmark compares the updating memory against a frozen-memory control and a volatility exposure gate under identical historical splits and costs (`FLY_GATE_CONTROL_RESULTS.json`). On a single daily BTC-USDC trajectory (720 bars, 5 seeds), `plastic fly` achieves +5.58% mean test return vs. +2.39% for `frozen` ($\Delta = +3.19\%$) and +3.26% for `linear ordinary` without the volatility gate, whereas under the static 80th-percentile volatility gate both `fly_gated` (−0.58%) and `frozen_gated` (−0.02%) are constrained on the single crypto trend path. Across 21 independent market trajectories (7 equity boards $\times$ 3 non-overlapping 3-year regimes, 2018–2026, 207,742 2D-balanced rows, $M = 5$ seeds) with complete-episode checked feedback ($G = \log(W_T/W_0)$), holding the 80th-percentile volatility gate, transaction fees (8 bps), and chronological splits strictly identical, `learn_with_gate` achieves a mean annualized Sharpe of $+1.71 \pm 0.14$ compared to -0.58 ± 0.11 for `frozen_with_gate` ($\Delta = +2.29 \pm 0.15$, paired $t = 8.36$, $p < 10^{-7}$, winning 21/21 trajectories), $+0.84 \pm 0.16$ for

`un-gated learn_no_gate`, and $+0.17 \pm 0.09$ for `dense linear ordinary_with_gate` (which cannot separate non-linear contrarian-vs.-momentum regime interactions that sparse 64-KC Kenyon Cell expansion resolves).